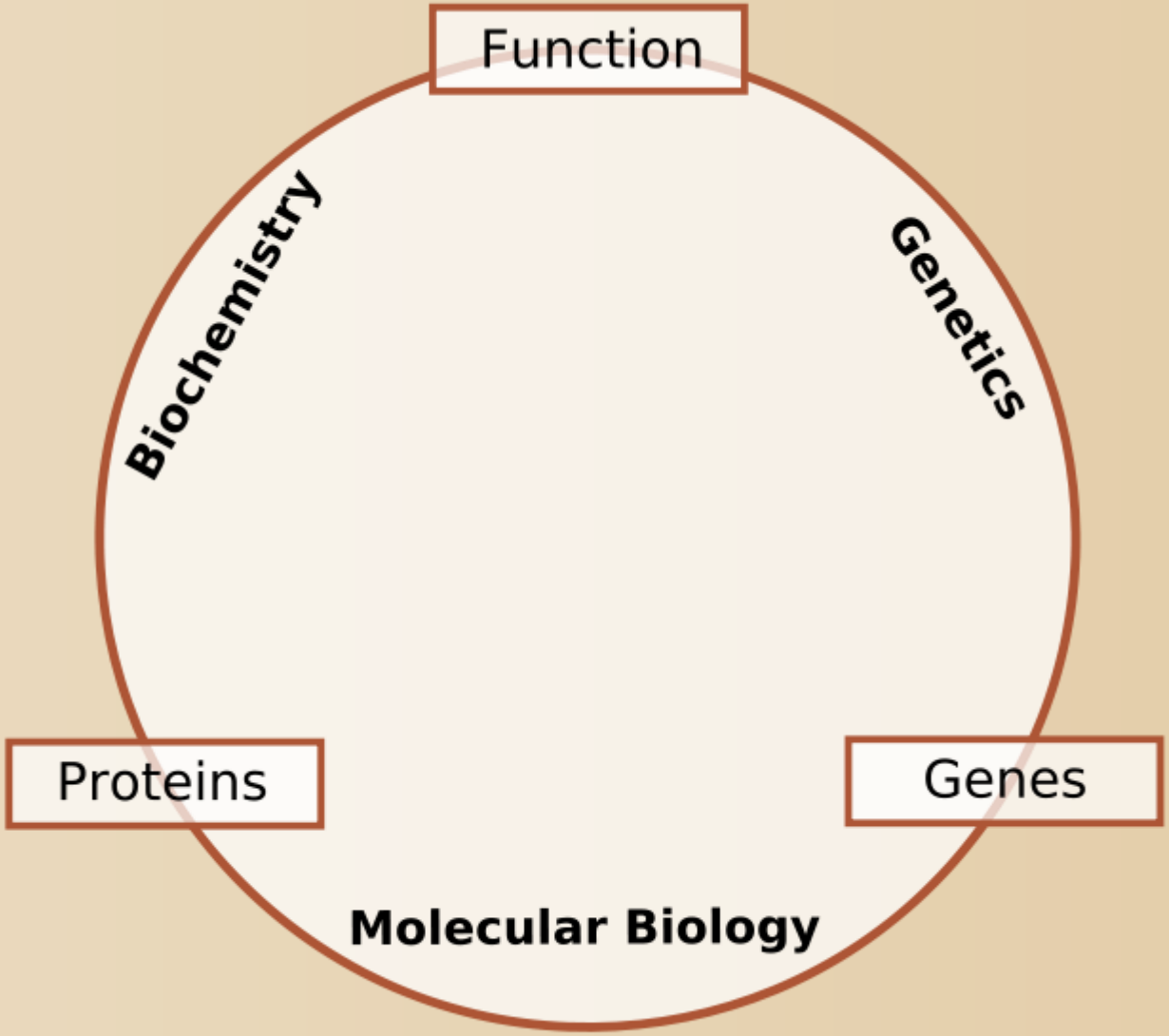
Modern Biology

1850 –



Before

In the 19 th Century, Darwin hypotized on variation, struggle for existence, and natural selection.

Key to the emergence of modern Biology in the 19th century was the development and refinement of the light microscope.

Cell theory and physiology, of course, led directly to the question of what happened when cells divided.

How do dividing cells contribute to the survival and evolution of the larger organism ?

Mendel (1822-1884)

Genetics was first studied scientifically by Gregor Mendel (an Austrian monk) in the 1860s.

Mendel grew and studied 29,000 pea plants from 1856 through 1863.

For his experiments, he selected the common pea plant, and in preliminary testing identified 7 pairs of characteristics for study. (Form, Color, Length, etc.)

Mendel carefully studied hybrids produced by crossing different varieties of pea plant. His was a pains-taking researcher that paid close attention to detail.

Mendel's Purpose

During that time, Mendel read On the Origin of Species and commented: "It requires some courage to undertake a labor for such far-reaching extent; this appears, however, to be the only right way by which we can finally reach the solution of a question the importance of which cannot be overestimated in connection with the evolution of organic forms."

Mendel was clearly aware of the larger issues of his great work and anticipated the impact his research would have on the theory of natural selection and evolution.

Mendel's Experiment

Mendel discovered that when he crossed one pure pea plant with the other, the hybrid tended to look exactly like one of the parent plants.

What Mendel observed was that some characteristics were transmitted virtually intact, with little change, while others seemed to disappear, also completely.

He called the former characteristics or traits dominant, and the latent characteristics or traits recessive.

https://www.youtube.com/watch?v=cWt1RFnWNzk

Gregor Mendel

Mendel's Experiment

The vocabulary of dominant and recessive characteristics is now commonplace, and we know that if we cross yellow and green peas, the first generation will have all yellow seeds because that trait is dominant.

Y=Yellow (dominate) and G = Green (recessive)

YY x gg

Yg Yg Yg Yg 1st Generation (All Yellow)

Yg x Yg

YY Yg YY gg 2nd Generation (3 Yellow, 1 Green)

Mathematics of Biology

Mendel experimented with all 7 of his traits and characteristics and determined that all 7 recessive traits reappeared in the same ratio (but not exact) of 1:3. He carried his experiments into several generations and determined that the ratios always remained the same.

Mendel showed that: “All constant combinations which in Peas are possible by the combination of the said seven independent differentiating characters were actually obtained by repeated crossing. Their number is given by 2^7 mathematical by the Laws of Combination

Mendel was probably the first biologist to work out a detailed application of mathematics to biology. Of course, he had no idea of the mechanism by which the Pea traits were transmitted.

Morgan's Research on Chromosomes

Morgan's group at Columbia University developed the technique of observing and mapping the changes in Drosophila (Common fruit fly) chromosomes under the microscope.

Model organisms are species that are studied to understand the biology of other organisms, often humans. Fruit flies share about 70% of the genes that cause disease with humans, so scientists can learn about human genetics by studying fruit fly genetics.

- They grow and multiply fast (short life cycle)

Morgan's Research on Chromosomes

This work led to the historic publication in 1915 of The Mechanism of Mendelian Heredity.

Morgan's research firmly established Mendelian principles with biologists.

https://www.youtube.com/watch?v=huDDaj0PjLU

Thomas Hunt Morgan and fruit flies

Darwin explained by Mendel

Experimental proof of Mendelian inheritance would have given importance support to evolutionary theory of Darwin.

The focus of genetics has been to seek an understanding of the biochemical basis of inheritance.

This field is now known as molecular biology.

Organic Compounds

In 1953, Stanley Miller and Harold Urey carried out an experiment at the University of Chicago where they mixed simple inorganic compounds that were thought to be present in the primordial Earth.

Miller & Urey mixed methane (CH_4), ammonia (NH_3), hydrogen gas (H_2), water (H_20), and carbon monoxide (CO) and applied electric arcs to the mixture to simulate lightning.

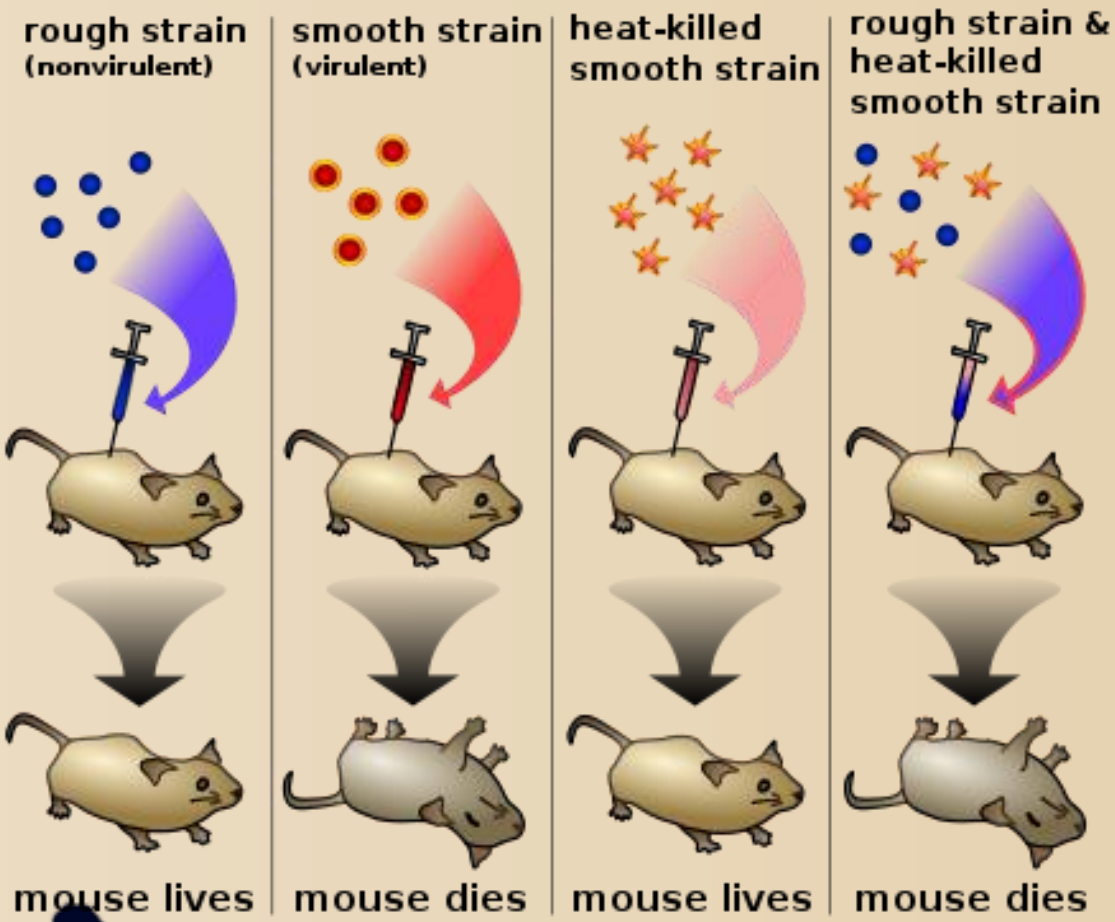
After one week, they found 10–15% of the carbon was present as organic compounds and amino acids had been created in the solution. Miller and Urey clearly demonstrated that the components of proteins could easily be produced from inorganic compounds under natural conditions.

Molecular Biology

Four landmark achievements have occurred in molecular biology. These were:

- The existence of DNA
- The elucidation of the structure of DNA
- The understanding of how DNA controls the structure of the proteins made by cells
- The discovery of the mechanism by which the DNA molecule replicates itself during cell division;

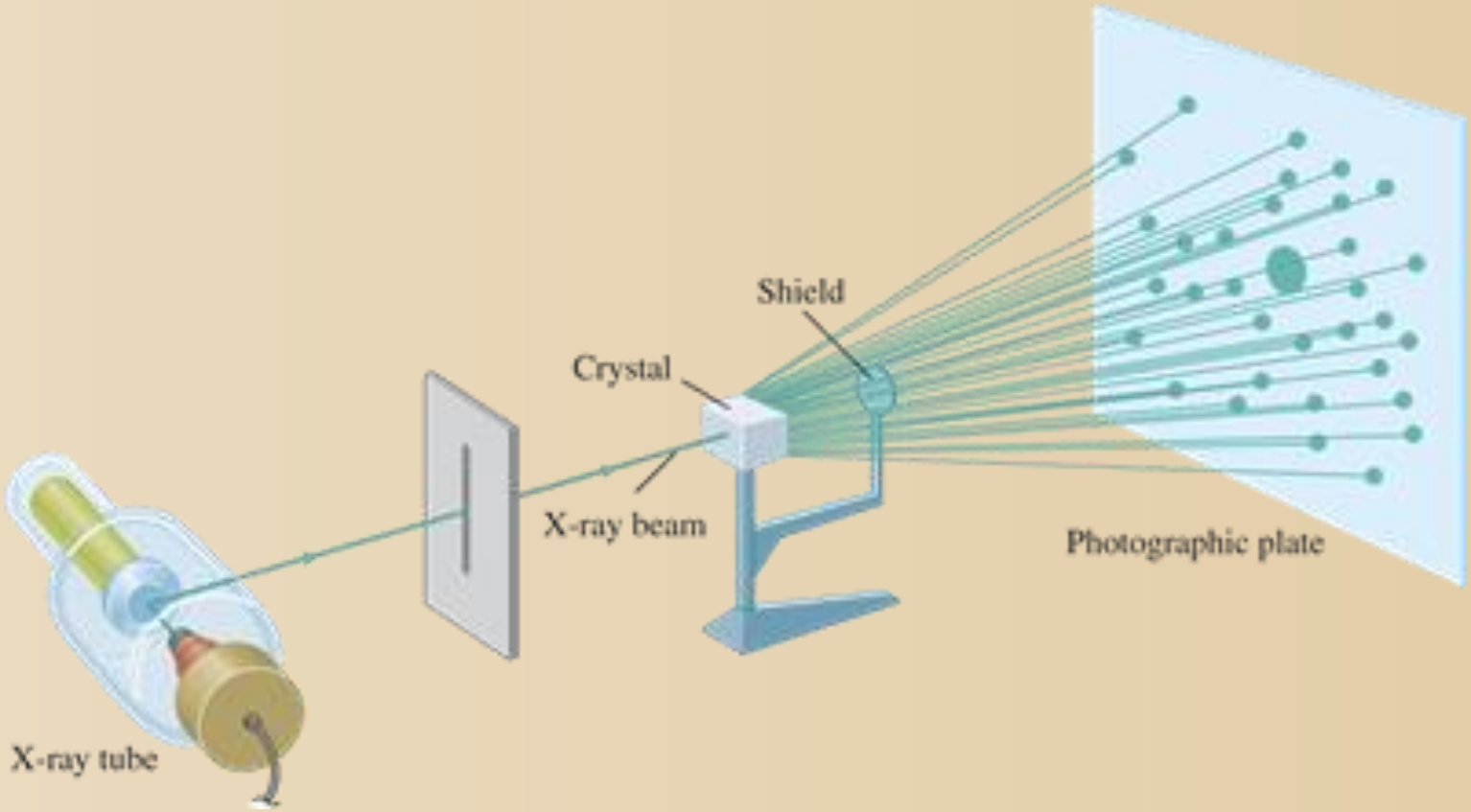
Griffith's experiment - 1928



The first experiment suggesting that bacteria can transfer genetic information through a process known as transformation.

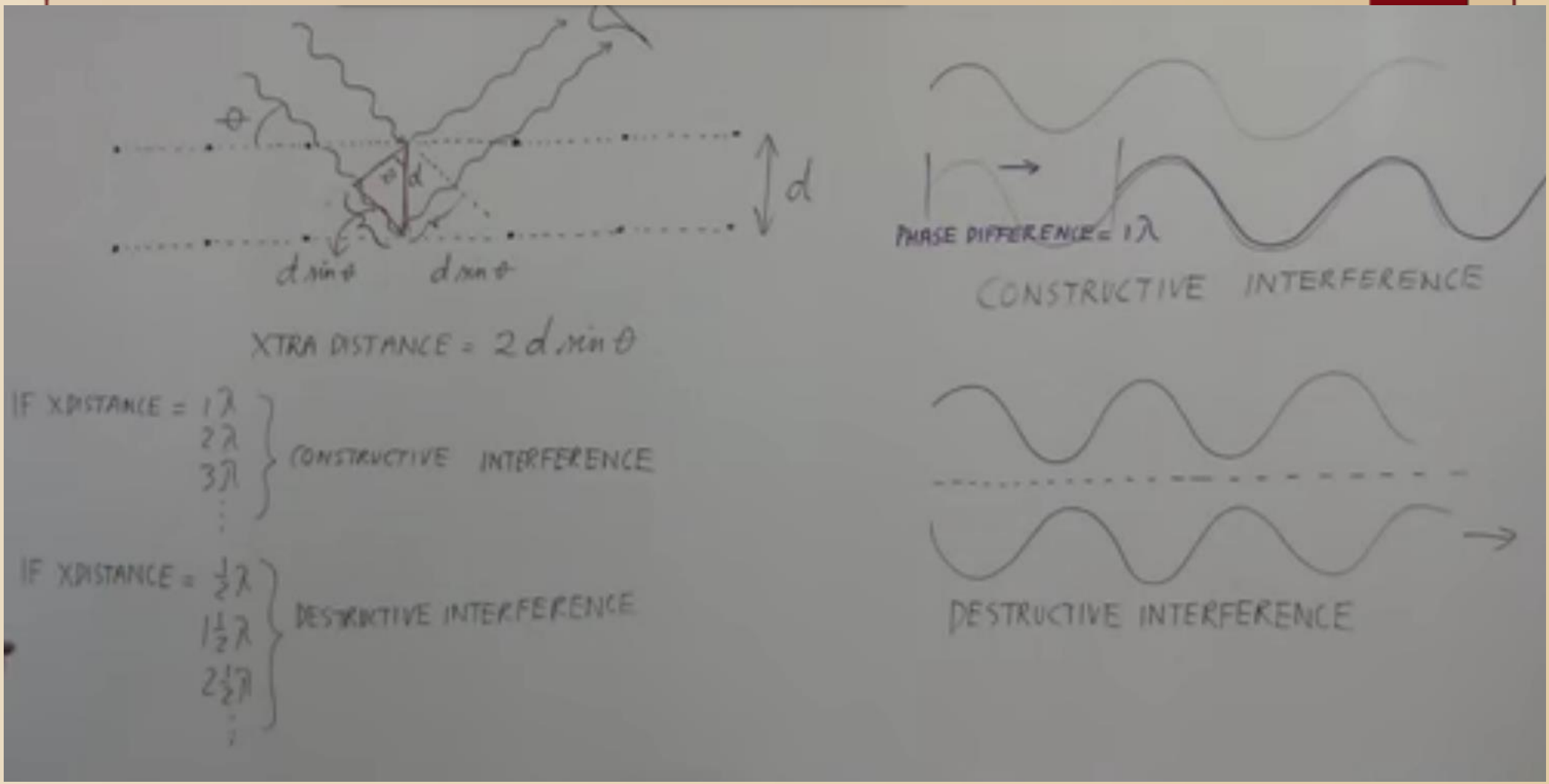
Griffith's findings were followed by research in the late 1930s and early 40s that isolated DNA as the material that communicated this genetic information.

X-Ray Crystallography



Bragg's (1912) studied the behavior of X-rays when shined on crystals. Atoms in crystals cause a beam of incident X-rays to diffract into many specific directions.

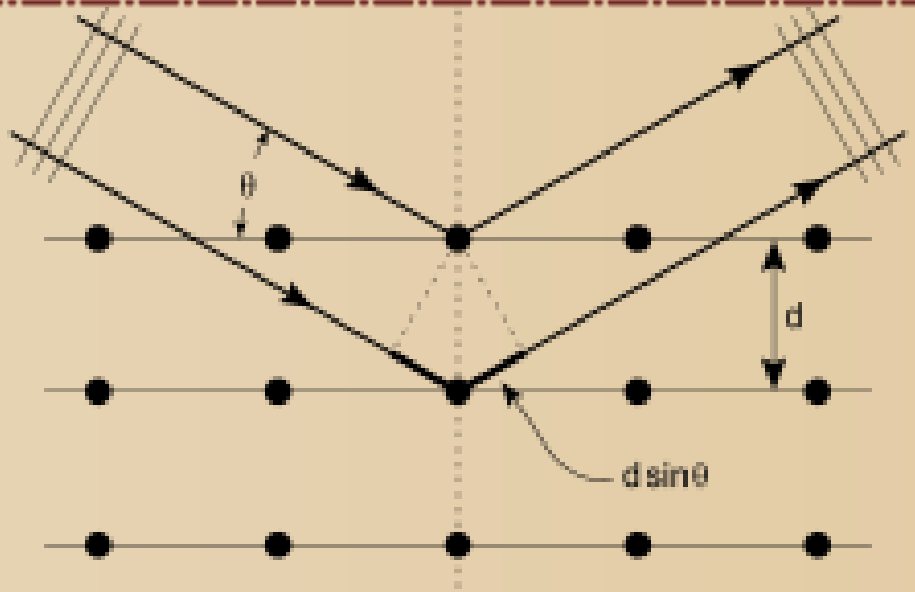
X-Ray Diffraction



Through a series of clever experiments, Bragg determined a mathematical relationship between the wavelength of the X-rays and the distance between the atoms in a crystal through reflection.

Crystal Structure

If scatterers are arranged symmetrically with a separation d, these spherical waves will be in sync (add constructively) only in directions where their path-length difference 2d sin θ equals an integer multiple of the wavelength λ.



$$n\lambda = 2d \sin\theta$$

where n is an integer, λ is the wavelength of X-rays, d is the distance between planes of a crystal and θ is the angle of diffraction.

Double Helix - DNA

A development in physics that was essential to the elucidation of the structure of DNA and its functioning was the invention of X-Ray Crystallography.

X-Ray_Diffraction_of_DNA

https://www.youtube.com/watch?v=u7RrXAjuNRk

The Discovery of the Structure of DNA

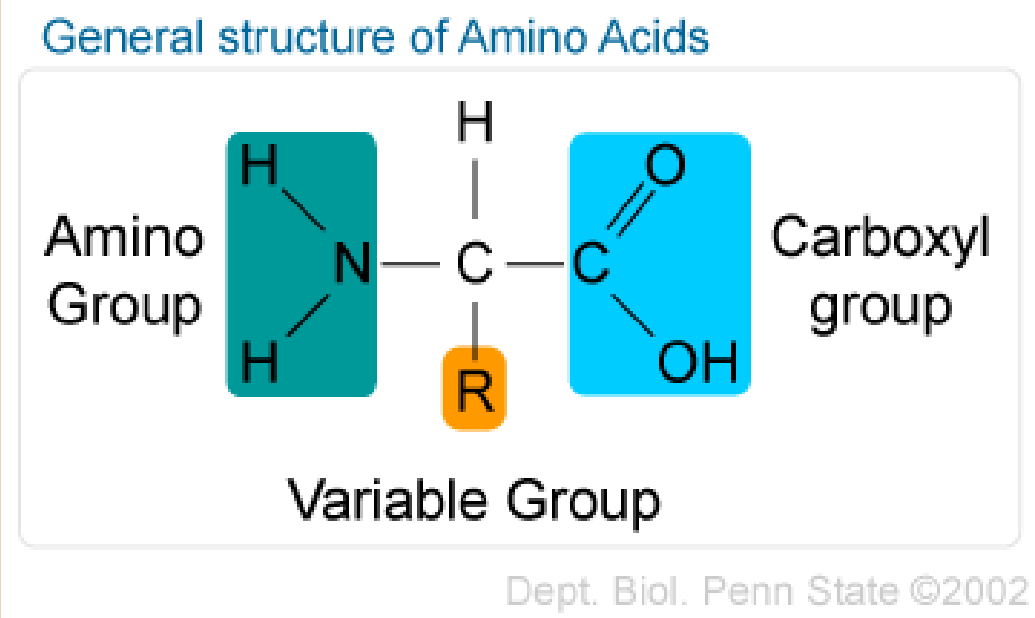
https://www.youtube.com/watch?v=V6bKn34nSbk

Amino Acids

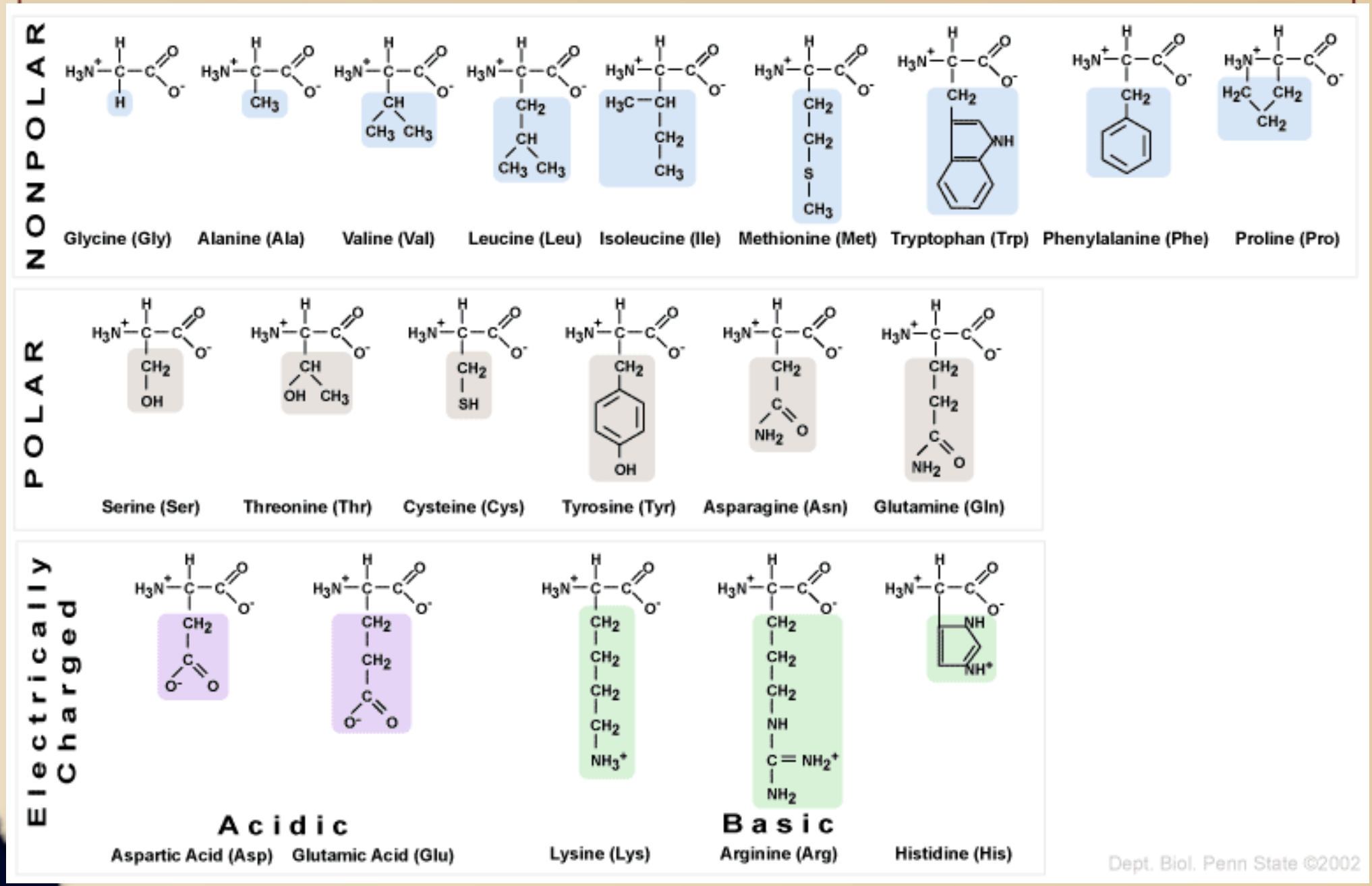
There are 20 kinds of amino acids found in nature and they all have both an amine end ($-NH_2$) and an acid end ($-CO_2H$).

The resulting molecule is still an amino acid having the characteristic amine and acid ends.

All amino acids have the same basic structure, an acid group at one end and an amine group at the other.



20 kind of Amino Acids



Proteins

Proteins are the building blocks of all living cells

They are very large molecules made up of a series of smaller molecules called amino acids. Amino acids are linked together to form long chains.

It is the order of the amino acids that determine the protein and its functionality.

Typical proteins of importance in the biochemistry of life are made up of hundreds or thousands of amino acids.

After the chains of Amino acids are constructed, they can fold into more complex structures.

Four Levels of Protein Structure

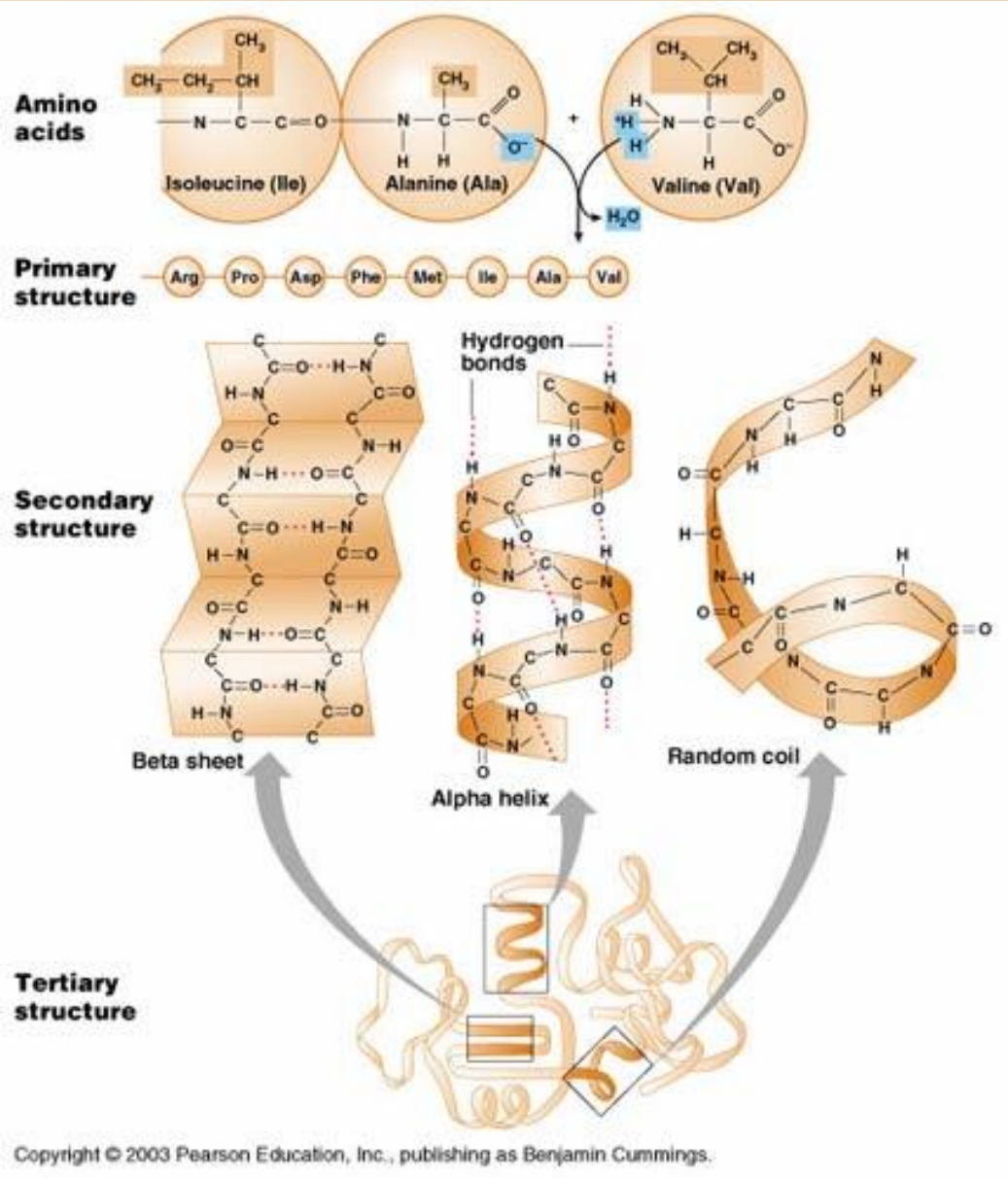
Primary - exact sequence of amino acids before folding.

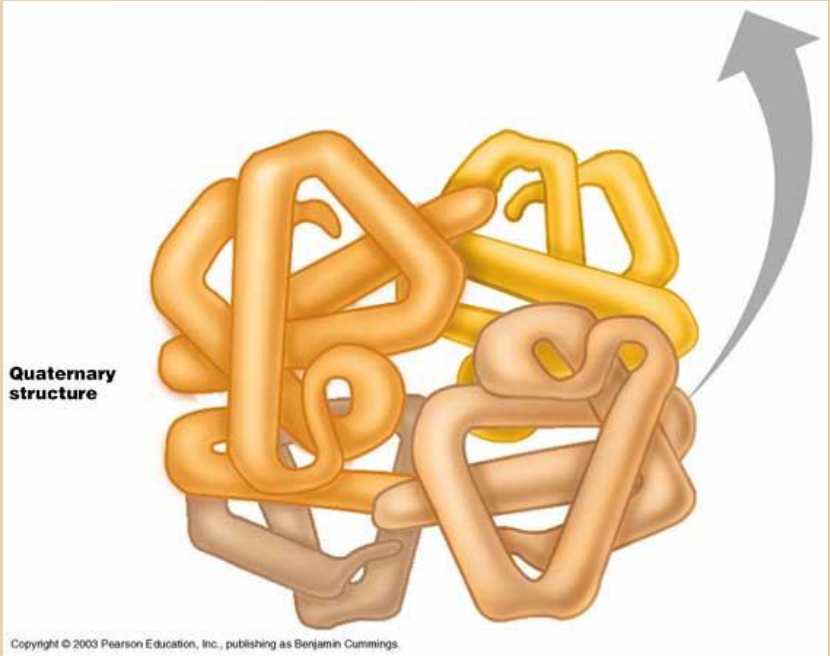
Secondary - simple folding create simple structures.

Tertiary - folding results in complex 3D structures.

Quaternary - multiple 3D subunits organized into a bigger structure.

Protein Structures





Genes (DNA) to Proteins

Genetic code was carried by four different bases in DNA, adenine, cytosine, guanine, and thymine.

It is the order of sets of three of these bases that allows DNA to select individual amino acids in the proper order when assembling proteins.

https://www.youtube.com/watch?v=gG7uCskUOrA

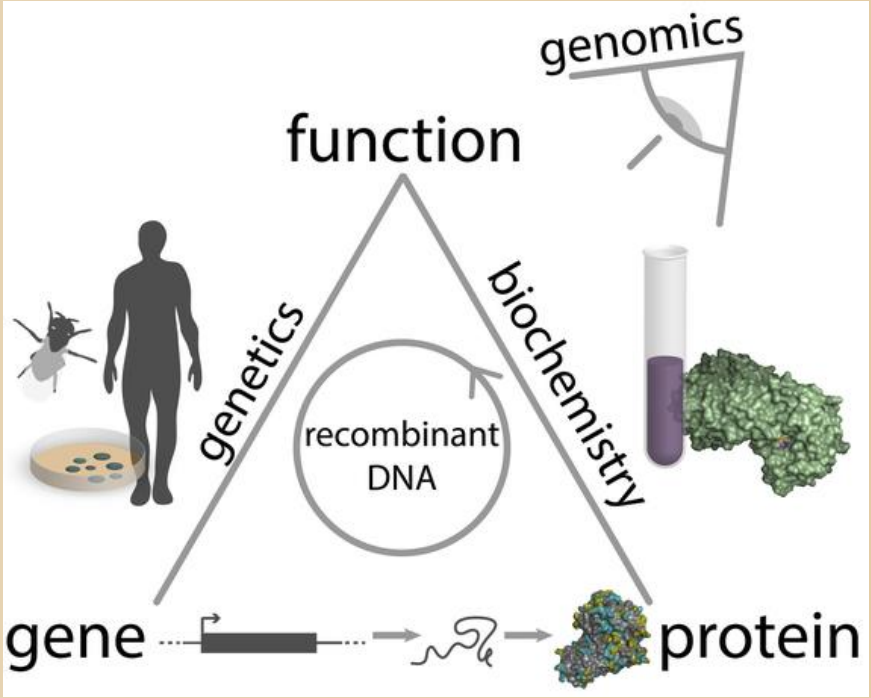
From DNA to protein - 3D

Central Dogma

Reference - Prof. Lander's Biology

Biochemistry focuses on the protein part of life functions. It studies the components independent of the organism.

Genetics focuses on the gene part. Usually mutants are used. So, it is organism without the component.



Molecular Biology integrates those two, as can be quite well ascertained from the "central dogma" i.e., genes-> proteins.

Study of Genomics

So, for e.g., if one is interested in studying what imparts red color to a fruit fly's eyes, this is probably how the three would work:

A biochemist would make a puree of the fruit fly, isolate the component responsible for the eye color and characterize it.

A geneticist would look for flies that have different eye colors, and compare each of them, breed them in various combinations, observe how the traits are inherited. So essentially, one can be blissfully unaware of the chemical nature of the said component (gene/ protein) but still figure out how the trait is passed on/ affect a population.

A molecular biologist would isolate the gene, study it, arrive at the protein therefrom